Project 2:

Iris plant classification using k-nearest neighbors

In [1]:
```
#import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
#exploring dataset
iris_df=pd.read_csv("/Users/mujeebmohammed/Downloads/ML Classification Pa
iris_df.head()
```

Out[2]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

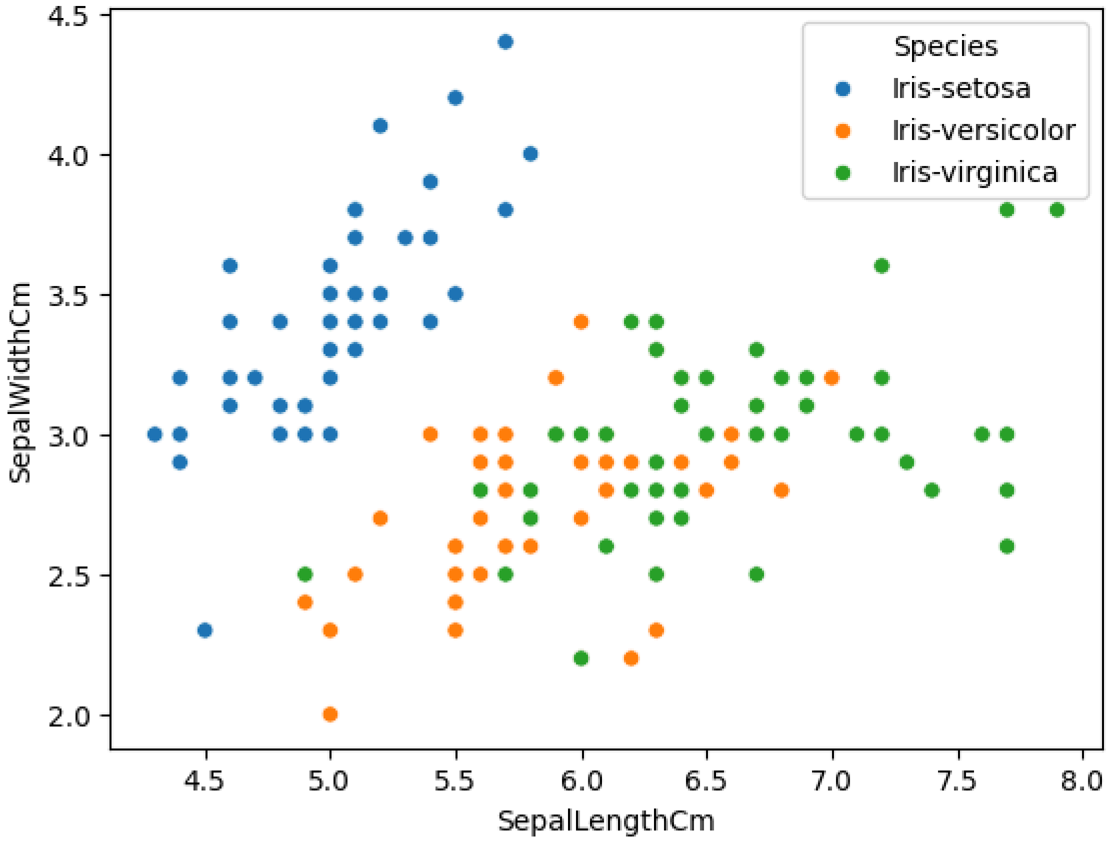
In [3]:
```
iris_df.tail()
```

Out[3]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

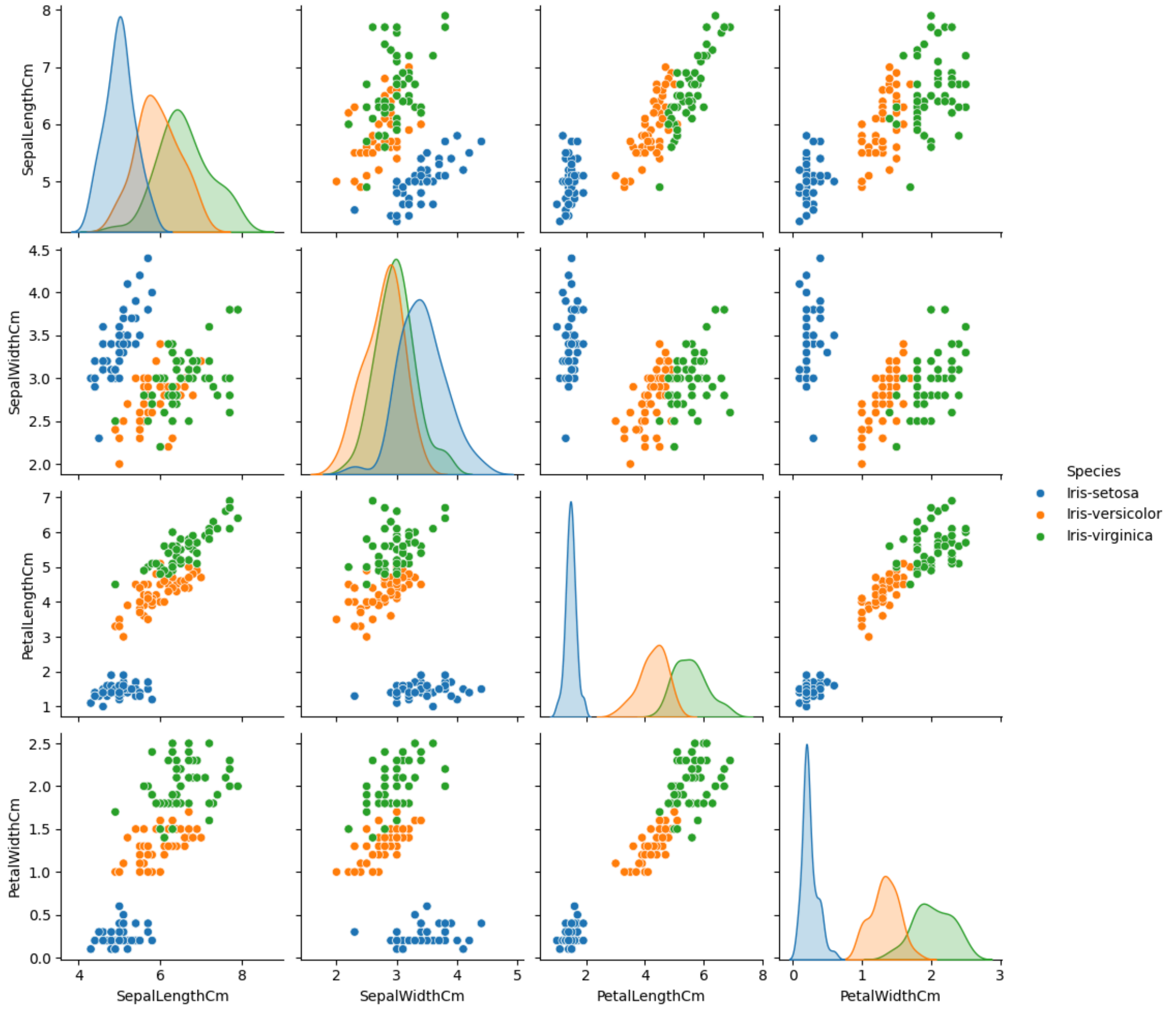
In [4]:
```
sns.scatterplot(x='SepalLengthCm',y='SepalWidthCm',hue='Species', data=ir
```

Out[4]:
```
<Axes: xlabel='SepalLengthCm', ylabel='SepalWidthCm'>
```



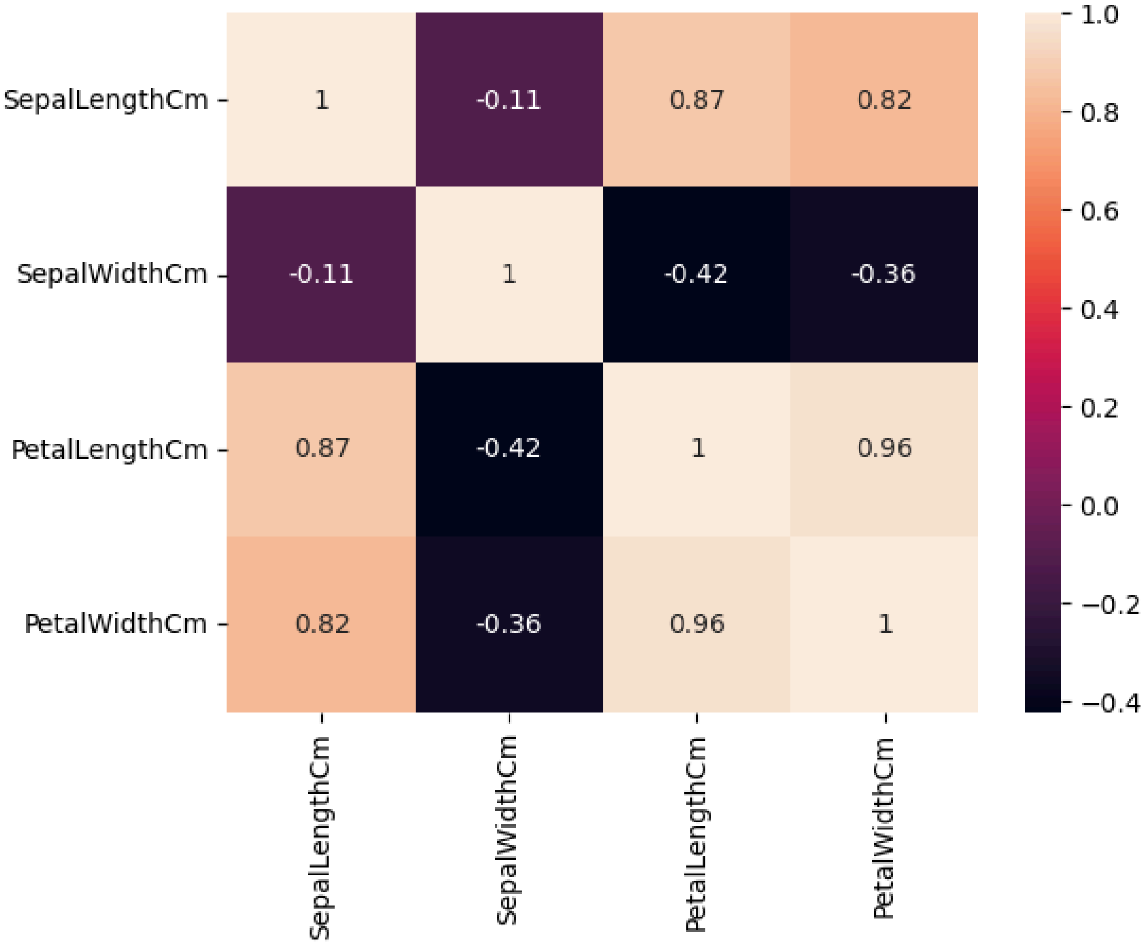
In [5]:
```
sns.pairplot(iris_df,hue='Species')
```

Out[5]: <seaborn.axisgrid.PairGrid at 0x125778050>



In [6]:
```
sns.heatmap(iris_df.select_dtypes(include='number').corr(),annot=True)
```

Out[6]: <Axes: >



In [7]:
```
#Data cleaning
X=iris_df.drop(['Species'],axis=1)
X
```

Out[7]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2
...	...	...	...	...
145	6.7	3.0	5.2	2.3
146	6.3	2.5	5.0	1.9
147	6.5	3.0	5.2	2.0
148	6.2	3.4	5.4	2.3
149	5.9	3.0	5.1	1.8

150 rows × 4 columns

In [8]:
```
y=iris_df['Species']
y
```

Out[8]:
```
0         Iris-setosa
1         Iris-setosa
2         Iris-setosa
3         Iris-setosa
4         Iris-setosa
             ...
145    Iris-virginica
146    Iris-virginica
147    Iris-virginica
148    Iris-virginica
149    Iris-virginica
Name: Species, Length: 150, dtype: str
```

In [9]:
```
from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.35)
```

In [10]:
```
X_train.shape
```

Out[10]:
```
(97, 4)
```

In [11]:
```
X_test.shape
```

Out[11]:
```
(53, 4)
```

In [12]:
```
X_train
```

Out[12]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
31	5.4	3.4	1.5	0.4
128	6.4	2.8	5.6	2.1
133	6.3	2.8	5.1	1.5
136	6.3	3.4	5.6	2.4
17	5.1	3.5	1.4	0.3
...	...	...	...	...
62	6.0	2.2	4.0	1.0
144	6.7	3.3	5.7	2.5
13	4.3	3.0	1.1	0.1
28	5.2	3.4	1.4	0.2
119	6.0	2.2	5.0	1.5

97 rows × 4 columns

In [13]:
```
#training the model
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import classification_report,confusion_matrix
```

In [14]:
```
classifier=KNeighborsClassifier(n_neighbors=5,metric='minkowski',p=2)
classifier.fit(X_train,y_train)
```

Out[14]:

▾ KNeighborsClassifier ⓘ ?

▸ Parameters

In [15]:

```
#evaluate
y_predict=classifier.predict(X_test)
y_predict
```

Out[15]:

```
array(['Iris-virginica', 'Iris-virginica', 'Iris-virginica',
       'Iris-setosa', 'Iris-virginica', 'Iris-versicolor',
       'Iris-versicolor', 'Iris-virginica', 'Iris-virginica',
       'Iris-setosa', 'Iris-virginica', 'Iris-setosa', 'Iris-setosa',
       'Iris-versicolor', 'Iris-setosa', 'Iris-versicolor',
       'Iris-virginica', 'Iris-virginica', 'Iris-setosa',
       'Iris-versicolor', 'Iris-virginica', 'Iris-virginica',
       'Iris-virginica', 'Iris-setosa', 'Iris-versicolor',
       'Iris-virginica', 'Iris-virginica', 'Iris-versicolor',
       'Iris-setosa', 'Iris-virginica', 'Iris-versicolor',
       'Iris-virginica', 'Iris-setosa', 'Iris-setosa', 'Iris-setosa',
       'Iris-virginica', 'Iris-setosa', 'Iris-virginica',
       'Iris-virginica', 'Iris-versicolor', 'Iris-virginica',
       'Iris-virginica', 'Iris-versicolor', 'Iris-versicolor',
       'Iris-virginica', 'Iris-virginica', 'Iris-setosa', 'Iris-setosa',
       'Iris-virginica', 'Iris-versicolor', 'Iris-setosa', 'Iris-setos
a',
       'Iris-versicolor'], dtype=object)
```

In [16]:

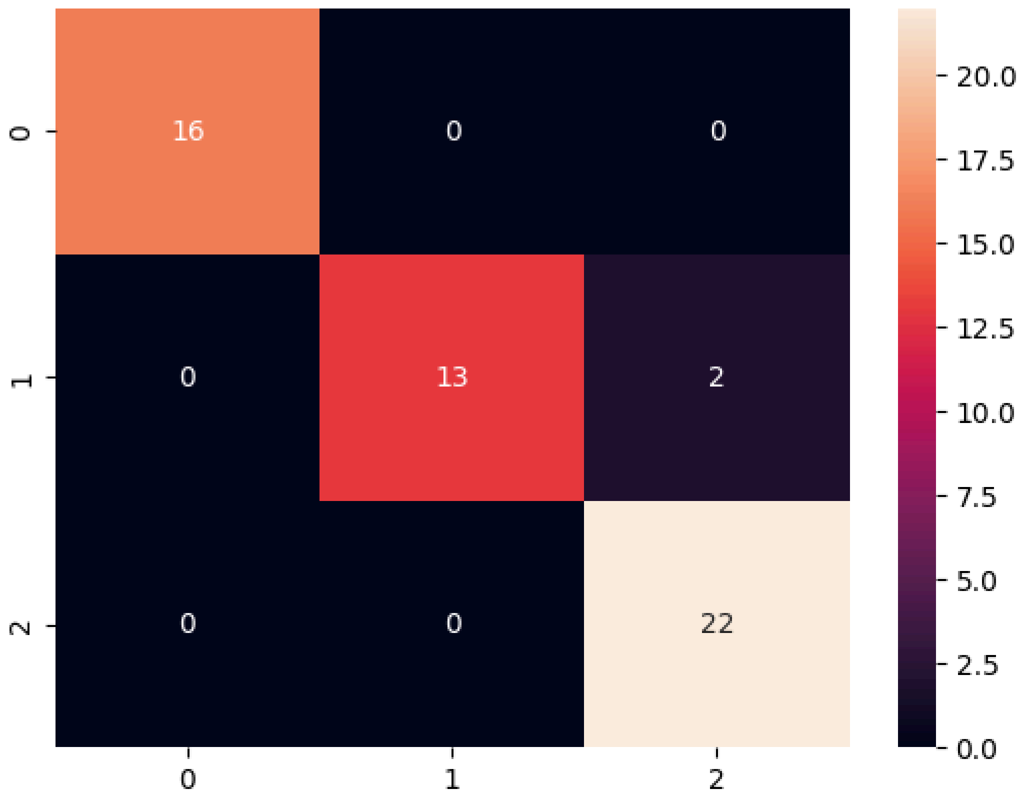
```
y_test
```

Out[16]:
```
83     Iris-versicolor
142     Iris-virginica
131     Iris-virginica
29         Iris-setosa
134     Iris-virginica
57     Iris-versicolor
53     Iris-versicolor
141     Iris-virginica
113     Iris-virginica
11         Iris-setosa
135     Iris-virginica
45         Iris-setosa
12         Iris-setosa
66     Iris-versicolor
0          Iris-setosa
99     Iris-versicolor
149     Iris-virginica
101     Iris-virginica
36         Iris-setosa
73     Iris-versicolor
116     Iris-virginica
105     Iris-virginica
143     Iris-virginica
37         Iris-setosa
71     Iris-versicolor
77     Iris-versicolor
148     Iris-virginica
79     Iris-versicolor
46         Iris-setosa
146     Iris-virginica
84     Iris-versicolor
114     Iris-virginica
27         Iris-setosa
9          Iris-setosa
19         Iris-setosa
132     Iris-virginica
8          Iris-setosa
121     Iris-virginica
139     Iris-virginica
95     Iris-versicolor
102     Iris-virginica
117     Iris-virginica
63     Iris-versicolor
61     Iris-versicolor
124     Iris-virginica
147     Iris-virginica
2          Iris-setosa
14         Iris-setosa
137     Iris-virginica
89     Iris-versicolor
21         Iris-setosa
43         Iris-setosa
76     Iris-versicolor
Name: Species, dtype: str
```

In [21]:
```
cm=confusion_matrix(y_test,y_predict)
```

In [22]:
```
sns.heatmap(cm,annot=True)
```

Out[22]: <Axes: >



In [23]:
```
print(classification_report(y_test,y_predict))
```

```
                 precision    recall  f1-score   support

    Iris-setosa       1.00      1.00      1.00        16
Iris-versicolor       1.00      0.87      0.93        15
 Iris-virginica       0.92      1.00      0.96        22

       accuracy                           0.96        53
      macro avg       0.97      0.96      0.96        53
   weighted avg       0.97      0.96      0.96        53
```